pfi

Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

Case Study: A Macro Agent on Two Statements

WEEK 2 · DAY 5

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Today: a macro agent on two statements

Week 2, working together

Day 6	market tools over the curve
Day 7	documents as tools, with sources
Day 8	tools behind a server
Day 9	the loop in a framework

Today

1. statement_diff: what changed, no model
2. The macro agent and its MacroNote
3. The answers decided first
4. The note: the step limit, then a fit
5. Every figure twice, every quote twice
6. Why did the curve move?
7. A workbook; what it shows, where it fails
8. Excel, Git, the AI check

Every model response in the notebook is a saved response, replayed. A live run can differ.

The rules for agents

On the statements: the wording, described; a changed word is a changed word

Allowed names

the statement states

a sentence in one and not the other

the date of the statement

Never called

hawkish or dovish

a policy view

what the Committee will do

On the curve: levels and changes in the file, never a view on rates

Allowed names

the 10 Yr par yield in the file

the change in bp between two dates

the slope in bp

Never called

a forecast or an outlook

a signal

a view on rates

statement_diff: what changed in the wording

2026-07-29

the earlier statement, from the file

2026-09-16

the later one, never typed

```
diff = agentkit.statement_diff(raw[FIRST], raw[SECOND])   # normalize_text first, then split
only in the first: 7; only in the second: 6
```

Why normalize first

The vote line as published holds U+2013, an en dash; normalize_text writes a hyphen with spaces.

Every row is a sentence as published: what the file says, not what it means.

How the splitter cuts

```
a sentence ends at ".", "?", or "!" followed by a space; a line break is a space
```

```
rows that begin with the vote line and run on into the next sentence: 2
rows that end at a single capital letter and a full stop: 2
the same statement with curly apostrophes: 0 rows
```

The vote line

ends with a colon, so it joins the decision

a new vote count changes the joined row

A person's initial

a full stop after it ends a "sentence"

the dissent sentence comes out in three pieces

Any list of changed sentences depends on where its sentences end.

The macro agent and its MacroNote

```
7 tools: ['statement_dates', 'statement_text', 'treasury_yield', 'curve_on', 'curve_change_bp', 'slope_bp', '...
```

```
macro_agent = agentkit.sdk_agent("macro desk", INSTRUCTIONS, list(macro_tools.values()), aclient,
                                 agentkit.MODELS["desk"], 10, output_type=MacroNote)
```

```
output type MacroNote: fields ['text', 'sentences_added', 'sentences_removed', 'figures', 'sources']
each figure: ['name', 'value', 'unit', 'tool_call_id', 'tenor']
store False, prompt_cache_key 'course8-day10'
```

CurveFigure subclasses agentkit.Figure and adds the tenor: check_figures reads a MacroNote as a DeskAnswer.

Every field description goes into the request, so each one is a prompt: no opinion word.

The case study: context, objective, data

Context

After each meeting, a desk writes a short note:

what the statement changed, and

what the Treasury curve did.

Objective

A note on the two statements and the curve

Every figure and every quote checked twice

Why did the curve move? Then a workbook

```
statements of 2026: ['2026-01-28', '2026-03-18', '2026-04-29', '2026-06-17', '2026-07-29', '2026-09-16']
2026-07-29 in the curve file: yes
2026-09-16 in the curve file: yes
2026-10-02 in the curve file: yes
2026-09-19 in the curve file: no
```

Course 7 read these statements as data on days 1 and 5. Today looks only at the rows the note touches.

The answers decided first

```
tenor      start        end  start_pct  end_pct  change_bp
 2 Yr 2026-07-29 2026-09-16       4.22     4.74       52.0
 2 Yr 2026-09-16 2026-10-02       4.74     4.83        9.0
10 Yr 2026-07-29 2026-09-16       4.67     5.01       34.0
10 Yr 2026-09-16 2026-10-02       5.01     5.28       27.0
```

```
2 Yr   2026-07-29 to 2026-09-16: the tool equals pandas: yes
2 Yr   2026-09-16 to 2026-10-02: the tool equals pandas: yes
10 Yr  2026-07-29 to 2026-09-16: the tool equals pandas: yes
10 Yr  2026-09-16 to 2026-10-02: the tool equals pandas: yes
```

What the file holds, decided before any request. No file of the course says why either yield moved.

The first form: the step limit

8

tool requests, then the limit

no

note returned

```
the run stopped at the step limit: Max turns (8) exceeded
tool requests: 8; {'statement_text': 2, 'treasury_yield': 6}
```

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 8 requests, 17,671 prompt and 224 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

Ten numbers one by one, plus two statements: twelve requests. curve_change_bp carries all ten in four.

The fix is not a larger limit. It is a question that names the tool and counts the requests.

The note, and its quotes against statement_diff

5
sentences added

5
removed

10
figures

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 7 requests, 15,322 prompt and 1,407 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

The note's quotes

every one in its statement, word for word

7 are rows of statement_diff

3 are cut where the diff did not cut

statement_diff's rows

7 quoted by the note, 6 not

5 of those: the same text, cut another way

1 left out: "Inflation remains elevated."

Two faithful lists can disagree. Check: is it in the statement word for word? Is it a change?

Every figure twice, and the note's own words

```
figures: 10; traced: 10; agree with pandas: 10; key numbers stated: 12 of 12
```

```
numbers in the note's text that no tool returned: []
```

```
assert_extraction_only: the text uses forecast, advice, or lending words: ['will']
```

Where the word came from

The note's own text restated a sentence of the later statement, in its own voice, without quotes.

The check cannot tell a restated sentence from a forecast; the note quotes, and describes.

A changed word is a changed word: the note may quote it, and never says what the Committee will do.

Why did the curve move?

```
WHY_QUESTION: by how many bp did the 10 Yr par yield change between the two dates, and why?
```

The number: from the tool

34 bp, from 4.67 to 5.01 percent

curve_change_bp's result, traced

The link: from no file

"likely contributed": the agent's sentence

the files hold no reason for a move

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 3 requests, 3,561 prompt and 324 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

```
numbers in the answer that no tool returned: []
```

Every number traces; the reason does not. The checks catch numbers. A reader catches reasons.

What the agent shows, and where it fails

```
sheets: ['figures', 'sentences']; rows [10, 16]
```

Shows	10 figures traced and agreeing; 10 quotes word for word
Fails: the limit	the first form listed ten numbers; no note
Fails: the lists	other sentence ends; the inflation line left out
Fails: own words	the Committee's word restated in the note's voice
Fails: why	a link to the decision that no file states

The curve moved between the dates; no file says why, and a note may not either.

Excel side by side, and git tag week2

```
=(INDEX(B2:O2941,MATCH(DATE(2026,9,16),A2:A2941,0),MATCH("2 Yr",B1:O1,0))
  -INDEX(B2:O2941,MATCH(DATE(2026,7,29),A2:A2941,0),MATCH("2 Yr",B1:O1,0)))*100
```

2 Yr change, 2026-07-29 to 2026-09-16	`Excel 52`
2 Yr change, 2026-09-16 to 2026-10-02	`Excel 9`
10 Yr change, 2026-07-29 to 2026-09-16	`Excel 34`
10 Yr change, 2026-09-16 to 2026-10-02	`Excel 27`
10 Yr change, 2026-07-29 to 2026-09-19	`Excel #N/A`
10 Yr, 2026-07-29 to 2026-09-16, without x 100	`Excel 0.34`

```
week2
(HEAD -> main, tag: week2) agentkit day 10: statement_diff, 52 tests
```

Excel 16.0 build 20430, typed by script; the workbook read back equal. 52 passed; the tag week2.

The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

Docstring

The later yield less the earlier, times 100: 1 percent is 100 bp.

2

Code back

Written for the course in the style of an AI answer. It runs.

3

Test first

Two hand rows, 4.00 and 4.25 percent. It returns 0.25 and fails.

4

Fix

The difference times 100: 25.0 on the rows, 34.0 on the file.

Today: curve_change_bp returns the change in percent and labels it bp: a 25 bp move prints as 0.25.

Code that runs is not the same as code that is right.

What you can do now

- statement_diff: the changed sentences, and how it cuts them
- A framework agent whose answer is your own MacroNote
- A question that fits the tools and the step limit
- Every figure and every quote checked twice
- No file says why the curve moved; read for that

Week 3, Day 11 An orchestrator and specialists